

```

.data
WordDecryptTestPhrase1: .asciiz "\nCurrent test case: "
WordDecryptTestPhrase2: .asciiz "\nResults are: "
SumNoGood: .asciiz " SUM result is WRONG"
CarryNoGood: .asciiz " CARRY result is WRONG"
AllGood: .asciiz " PASSES TEST"

```

```

WordDecryptTestData: .word 100, 200, 1, 301, 0
.word 60000, 30000, 1, 90001, 0
.word 600000, 300000, 1, 900001, 0
.word 2147483640, 2147483640, 0, -16, 0
.word -100, 5000, 0, 4900, 1
.word 50468, -1248128, 1, -1197659, 0
.word 50468, -30303, 0, 20165, 1
.word 0x0, 0x0, 0x0

```

```

.text

```

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main:

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    la $t0, WordDecryptTestData
WordDecryptTestLoop:
    la $a0, WordDecryptTestPhrase1
    li $v0, 4
    syscall
    lw $a0, 0($t0)
    li $v0, 1
    syscall
    li $a0, '+'
    li $v0, 11
    syscall
    lw $a0, 4($t0)
    li $v0, 1
    syscall
    li $a0, '+'
    li $v0, 11
    syscall
    lw $a0, 8($t0)
    li $v0, 1
    syscall

    lw $a0, 0($t0)
    lw $a1, 4($t0)
    lw $a2, 8($t0)
    sw $t0, -4($sp)
    addi $sp, $sp, -4
    jal WordDecrypt

    lw $t0, 0($sp)
    addi $sp, $sp, 4
    move $t1, $v0 # store v0 in t1
    move $t2, $v1 # store v1 in t2

    li $a0, '='
    li $v0, 11
    syscall
    move $a0, $t1
    li $v0, 1
    syscall
    li $a0, ', '
    li $v0, 11
    syscall
    move $a0, $t2

```

```
li $v0,1
syscall
```

```
li $t5, 1
lw $t3, 12($t0)
lw $t4, 16($t0)
beq $t1, $t3, SUMGOOD
li $t5, 0
la $a0, SumNoGood
li $v0, 4
syscall
```

SUMGOOD:

```
beq $t2, $t4, CARRYGOOD
li $t5, 0
la $a0, CarryNoGood
li $v0, 4
syscall
```

CARRYGOOD:

```
beq $t5, $zero, SKIPPEDPASS
```

```
la $a0, AllGood
li $v0, 4
syscall
```

SKIPPEDPASS:

```
addi $t0, $t0, 20
lw $t1, 0($t0)
bne $t1, $zero, WordDecryptTestLoop
```

EndWordDecryptTest:

```
li $v0, 10
syscall
```

Insert your WordDecrypt Here